

# Real Estate Sales & Booking Analytics Dashboard

Power BI Capstone Project — Report

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## Project Title & Objective

**Objective:** Build a professional Power BI dashboard for Real Estate Sales & Booking Analytics. The solution includes data cleaning, star schema modeling, DAX calculations, interactive dashboards, and business insights to support management decisions.

## Dataset Used

Bookings (Fact Table), Customers (Dimension), Locations (Dimension), Date Table. Data includes bookings, customers, project details, locations, payments, configuration, booking status, and dates.

## Data Cleaning & Transformation

- Removed duplicate Booking\_ID and Customer\_ID.
- Trimmed and cleaned text columns.
- Standardized text casing.
- Corrected data types.
- Handled missing values logically.
- Removed invalid records (zero price, negative discounts).
- Built relationships and star schema (Bookings ↔ Customers ↔ Locations).
- Created calculated columns (Booking Year, Month, Quarter, Age Group, Days to Possession, Payment Completion %, Price Category, Loan Flag, Discount Category, etc.).

## Visualizations & Dashboards

### Executive Overview

- KPI Cards (Total Sales, Total Bookings, Total Customers, Collection %, Balance Due, Avg Booking Value)
- Monthly Sales Trend
- Total Sales by City
- Booking Status Donut
- Interactive Slicers (Year, Month, City, Booking Status, Configuration)

### Project & Location Analysis

- Configuration-wise Bookings (Donut)
- Average Price per Sqft by Project
- Total Properties by Project Status
- Top 10 Projects by Sales
- Total Sales by City and State (Map)

## Business Insights

- Highest revenue city identified — Faridabad leads total sales among tracked cities.
- Top-performing projects by sales (DLF The Mer..., ATS Metro H..., Supertech Gr... lead the ranking).
- Configuration booking distribution is fairly balanced across 2BHK, 3BHK, 4BHK, 1BHK, Studio, Villa, Plot and Penthouse (each roughly 11-14% share).
- Average price per square foot varies significantly by project, useful for pricing benchmarking.
- Project status distribution shows a near-even split across Under Construction, Launched, Nearing Completion and Ready to Move.
- Geographic sales concentration visualized using the City/State map (Delhi, Haryana, Uttar Pradesh).
- Sales trend over time tracked using the Date table and time-intelligence measures.

## Challenges & Solutions

- Incorrect data types (dates/numbers stored as text) — resolved in Power Query using explicit type conversion.
- Built star schema relationships between Bookings, Customers and Locations to avoid many-to-many ambiguity.
- Resolved DAX errors using VALUE(), FORMAT(), and proper data typing before aggregation.
- Organized measures into a dedicated measure table for cleaner model navigation.

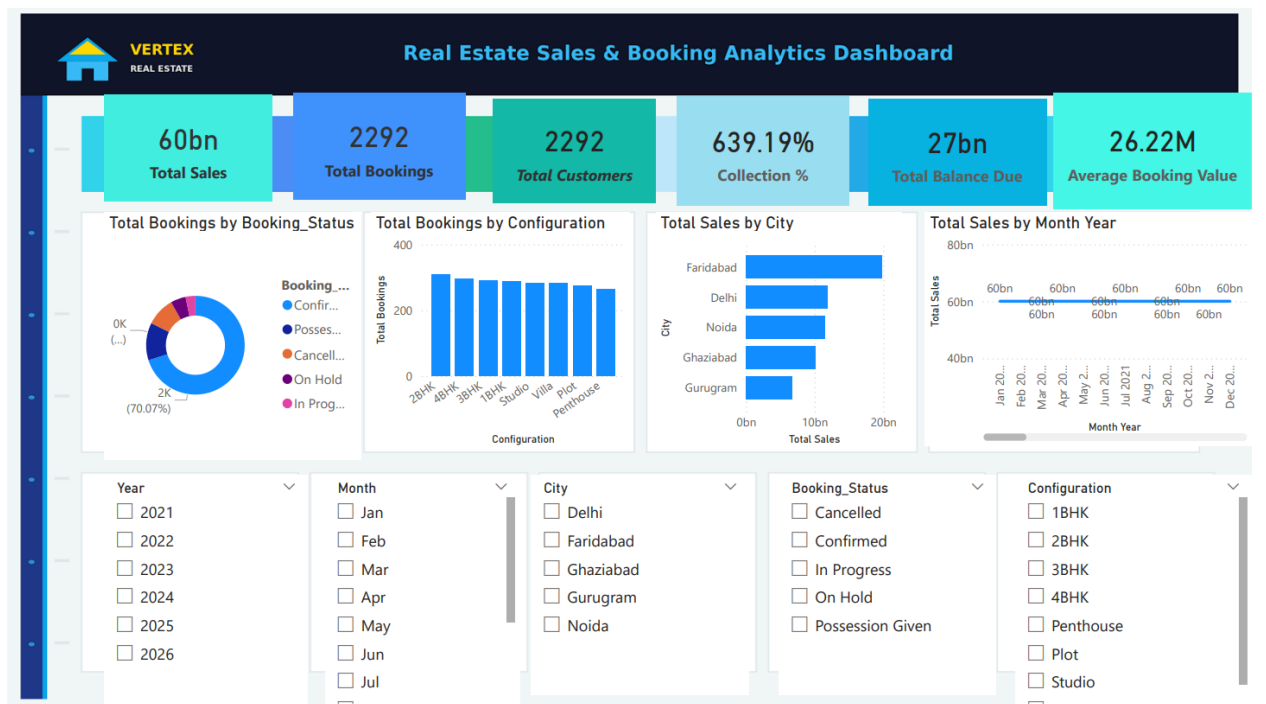
## Future Improvements

- Drill-through pages (City → Project → Booking detail).
- Dynamic tooltips with mini charts.
- Sales forecasting using built-in forecast lines.
- AI visuals (Key Influencers, Decomposition Tree).
- Row-Level Security (RLS) for regional sales teams.
- Bookmarks and page navigation buttons for a guided report experience.

## Conclusion

This project demonstrates a complete Power BI analytics solution — from raw data cleaning to a professional, branded, interactive dashboard with DAX measures and business-ready insights suitable for executive reporting.

## Executive Overview Dashboard



## Project & Location Analysis Dashboard

